



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 2)

2020 HSC Course Assessment Task 1

11th of December, 2019

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.
- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER:

BOOKLETS USED:

Class (please ✓)

- ☐ 12M4A – Mr Lin
- ☐ 12M4B – Dr Jomaa
- ☐ 12M4C – Ms Ziazaris

Marker's use only.

QUESTION	1	2	3	4	5	6	Total	%
MARKS	10	8	7	5	6	7	43	

Write your answers in the writing booklet supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 1 (10 Marks)	Commence a NEW booklet.	Marks
(a) Given $z = -1 + i\sqrt{3}$ and $\omega = 1 + i$.		
i. Evaluate $ z $.		1
ii. Find $\frac{\omega}{z}$ in the $x + iy$ form, where x and y are real numbers.		2
iii. Find $\omega\bar{z}$ in modulus argument form.		2
(b) Given $z = x + iy$, if $ z - 3 - 4i = 4$, find z such that $x^2 + y^2$ is minimum.		2
(c) Solve for z , such that $z^2 + \frac{1}{z^2} = 1$. Express your answers in the $x + iy$ form.		3

Question 2 (8 Marks)	Commence a NEW booklet.	Marks
(a) Write the converse of the following statement: If quadrilateral $ABCD$ is a square, then quadrilateral $ABCD$ is a rectangle.		1
(b) Write the contrapositive of the following statement: If two distinct lines in the xy plane are parallel, then they have the same slope.		1
(c) Negate the following statement: If Harold passes his maths course and finishes his statistics project, then he will graduate at the end of the year.		1
(d) Write the following statement in symbolic form: There exists an even integer divisible by 5.		1
(e) Prove that the following statement is false, giving a counter example: If a positive integer p is prime, then so is $2p + 1$.		1
(f) Prove that if a positive integer n has remainder 1 when divided by 3, then n^3 has remainder 1 when divided by 3.		3

Question 3 (7 Marks)

Commence a NEW booklet.

Marks

- (a) A polynomial function is

$$P(x) = x^4 - 5x^3 + 7x^2 + 3x - 10.$$

- i. Show that
- $2 - i$
- is a root of
- $P(x)$
- .

1

- ii. Hence, or otherwise, factorise
- $P(x)$
- completely over
- $\mathbb{C}$
- .

3

- (b) Sketch the region in the Argand diagram consisting of those point
- z
- for which

3

$$|z - 2 + i| > |z + 2 + 2i| \quad \text{intersect with} \quad 0 \leq \arg(z + 2 + 2i) \leq \frac{\pi}{4}.$$

Show all the important features.

Question 4 (5 Marks)

Commence a NEW booklet.

Marks

- (a) Prove (without using induction) that for all integers
- $n \geq 2$

2

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \cdots \left(1 - \frac{1}{(n-1)^2}\right) \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}.$$

- (b) Prove by contradiction the following:

3

$$(\forall x \in \mathbb{Z})(\forall y \in \mathbb{Z})[(3x + \sqrt{2}y = 0) \implies (x = 0 \text{ and } y = 0)]$$

Question 5 (6 Marks)

Commence a NEW booklet.

Marks

Given that

$$(x + y)^5 - (x^5 + y^5) = 5xy(x + y)(x^2 + xy + y^2).$$

and

$$z^n + z^{-n} = 2 \cos n\theta.$$

- i. Show that

2

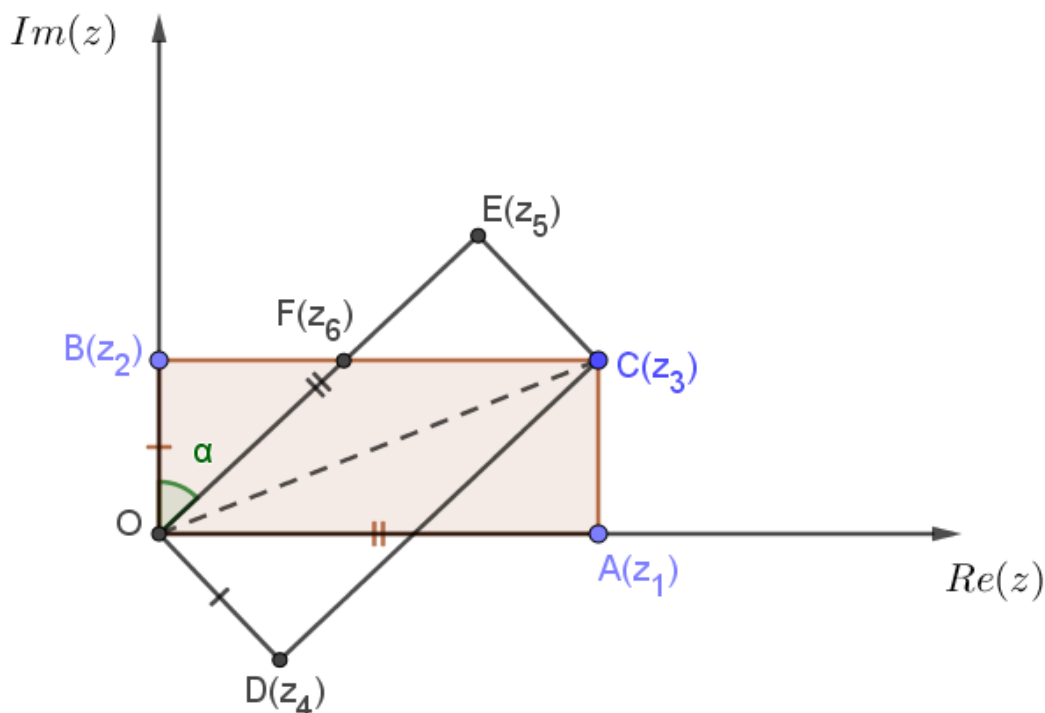
$$\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta.$$

- ii. Hence, show that the equation
- $16x^5 - 20x^3 + 5x - 1 = 0$
- has roots
- 2**
-
- $1, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \text{ and } \cos \frac{8\pi}{5}.$

- iii. Use (ii) to show that the equation
- $16x^4 + 16x^3 - 4x^2 - 4x + 1 = 0$
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Given z_1 and z_2 two complex numbers in the Argand diagram (see diagram).

- i. Find z_3 in terms of z_1 and z_2 , such that $OACB$ formed a rectangle. 1
- ii. Find z_4 and z_5 in terms of z_1 , z_2 and α , where $\alpha = \angle EOB$. 2
- iii. Hence, or otherwise show that 2

$$z_3 = i(z_1 - z_2)\overline{\csc \alpha}.$$

- iv. The complex number z_6 represents the point F (the intersection of BC and OE). Show that 2

$$z_6 = \frac{(z_1 + z_2)}{2} \left(1 + i \cot \left(\frac{\pi + 2\alpha}{4} \right) \right).$$

End of paper.

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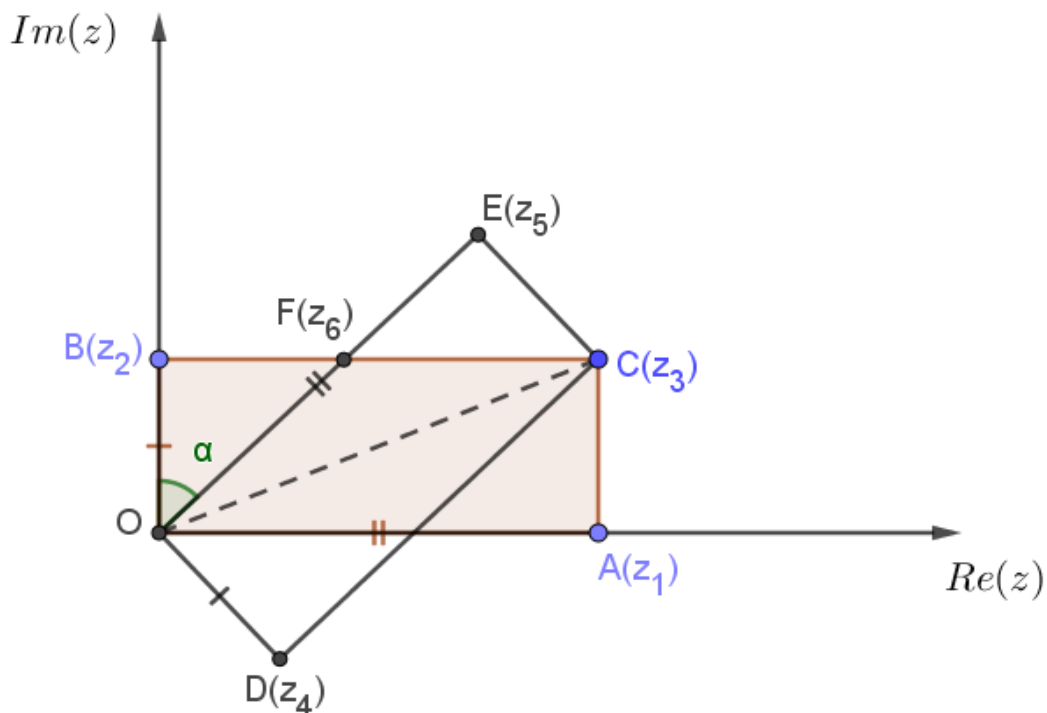
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End of paper.

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Suggested Solutions

Section II

Question 1

(a) $z = -1 + i\sqrt{3}$ and $\omega = 1 + i$.

i. $|z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2$.

ii. $\frac{\omega}{z} = \frac{1+i}{-1+i\sqrt{3}} = \frac{(1+i)(-1-i\sqrt{3})}{(-1+i\sqrt{3})(-1-i\sqrt{3})} = \frac{-1-i\sqrt{3}-i+\sqrt{3}}{4} = \frac{-1+\sqrt{3}}{4} - \frac{1+\sqrt{3}}{4}i$.

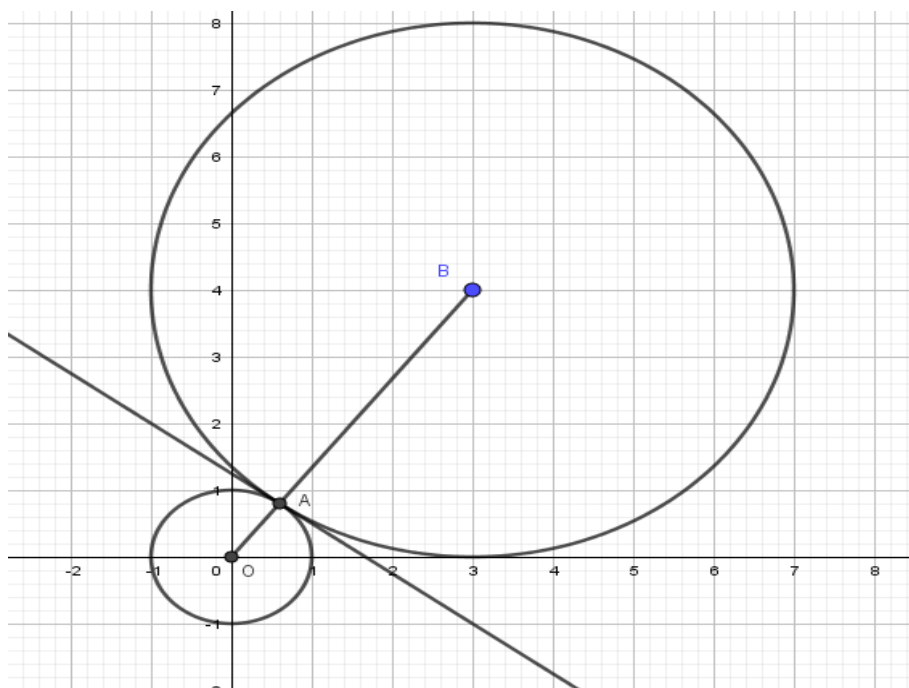
iii.

$$\omega = 1 + i = \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right) = \sqrt{2} \operatorname{cis} \frac{\pi}{4}.$$

$$\bar{z} = -1 - i\sqrt{3} = 2 \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right) = 2 \operatorname{cis} \frac{4\pi}{3}.$$

$$\omega \bar{z} = \left(\sqrt{2} \operatorname{cis} \frac{\pi}{4} \right) \left(2 \operatorname{cis} \frac{4\pi}{3} \right) = 2\sqrt{2} \operatorname{cis} \left(\frac{\pi}{4} + \frac{4\pi}{3} \right) = 2\sqrt{2} \operatorname{cis} \frac{19\pi}{12} = 2\sqrt{2} \operatorname{cis} \left(\frac{-5\pi}{12} \right).$$

- (b) The circle $(x-3)^2 + (y-4)^2 = 16$ centered at $(3, 4)$ and has radius 4. The distance from the origin to the center equal 5 (by using Pythagoras theorem) and the shortest distance to the circle is $5 - 4 = 1$. The line through the origin and the center has equation $4x - 3y = 0$. The x -value of the points of intersection of this line and the circle are given by $(x-3)^2 + \left(\frac{4x}{3} - 4\right)^2 = 16$ which give $x = \frac{3}{5}$ and $x = \frac{27}{5}$. Hence $y = \frac{4}{5}$ and $y = \frac{36}{5}$. So $z = \frac{3}{5} + \frac{4}{5}i$ and $|z| = 1$.



$$(c) \quad \left(z + \frac{1}{z}\right)^2 = z^2 + 2 + \frac{1}{z^2} = z^2 + \frac{1}{z^2} + 2 = 1 + 2 = 3.$$

$$z + \frac{1}{z} = \pm\sqrt{3}.$$

z are the solutions of $z^2 + \sqrt{3}z + 1 = 0$ and $z^2 - \sqrt{3}z + 1 = 0$.

$$b^2 - 4ac = 3 - 4 = -1 = i^2.$$

$$\text{Hence } z = \frac{\sqrt{3}}{2} + \frac{1}{2}i, \quad \frac{\sqrt{3}}{2} - \frac{1}{2}i, \quad \frac{-\sqrt{3}}{2} + \frac{1}{2}i, \quad \frac{-\sqrt{3}}{2} - \frac{1}{2}i.$$

Question 2 solutions

- (a) **Converse:** If Quadrilateral $ABCD$ is a rectangle, then the quadrilateral $ABCD$ is a square.
- (b) **Contrapositive:** If two distinct lines in the xy plane do not have the same slope, then they are not parallel.
- (c) **Negation** :Harold passes his math course and he finishes his statistics project, but he did not graduated at the end of the year.
- (d) **Symbolic form:** $(\exists p \in \mathbb{Z})(\exists m \in \mathbb{Z})(2p = 5m)$.
- (e) **Counter example:** 7 is a prime number but $15 = 2 \times 7 + 1$ is not prime.
- (f) Let n be a positive integer which has remainder 1 when divided by 3, ie. $\exists m \in \mathbb{Z}, n = 3m + 1$. We need to prove that n^3 has remainder 1 when divided by 3.

$$n^3 = (3m+1)^3 = (3m)^3 + 3(3m)^2 + 3(3m) + 1 = 27m^3 + 27m^2 + 9m + 1 = 3(9m^3 + 9m^2 + 3m) + 1 = 3k + 1,$$

where $k = 9m^3 + 9m^2 + 3m \in \mathbb{Z}$. Hence the remainder of dividing n^3 by 3 is 1.

Question 3 solutions

(a) $P(x) = x^4 - 5x^3 + 7x^2 + 3x - 10$.

i.

$$(2 - i)^2 = 4 - 4i + i^2 = 3 - 4i$$

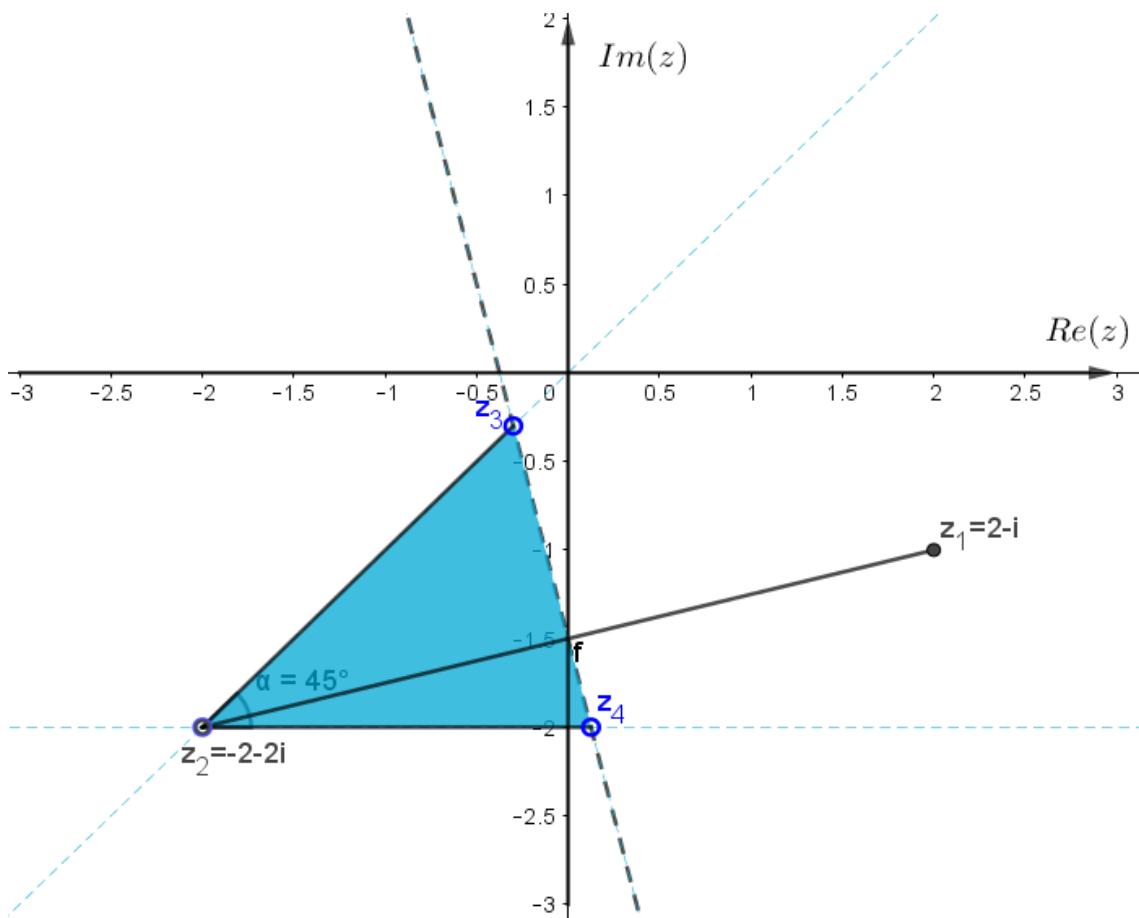
$$(2 - i)^3 = (3 - 4i)(2 - i) = 6 - 3i - 8i + 4i^2 = 6 - 4 - 11i = 2 - 11i$$

$$(2 - i)^4 = (2 - 11i)(2 - i) = 4 - 2i - 22i + 11i^2 = 4 - 11 - 24i = -7 - 24i$$

$$\begin{aligned} P(2 - i) &= (2 - i)^4 - 5(2 - i)^3 + 7(2 - i)^2 + 3(2 - i) - 10 = \\ &= -7 - 24i - 5(2 - 11i) + 7(3 - 4i) + 3(2 - i) - 10 = \\ &= -7 - 24i - 10 + 55i + 21 - 28i + 6 - 3i - 10 = 0 + 0i = 0 \end{aligned}$$

ii. $P(x)$ has integer coefficients and $2 - i$ is a root implies that $2 + i$ is also a root. So $(x - 2 + i)$ and $(x - 2 - i)$ are factors of $P(x)$. Hence $x^2 - 4x + 5 = (x - 2 + i)(x - 2 - i)$ is a factor of $P(x)$. Using long division or alternative approach, you can show that $P(x) = (x^2 - 4x + 5)(x^2 - x - 2) = (x - 2 + i)(x - 2 - i)(x - 2)(x + 1)$.

(b) $z_1 = 2 - i$ and $z_2 = -2 - 2i$



Question 4 solutions

(a)

$$\begin{aligned}
& \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \cdots \left(1 - \frac{1}{(n-1)^2}\right) \left(1 - \frac{1}{n^2}\right) = \\
& \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \cdots \left(1 - \frac{1}{(n-1)}\right) \left(1 - \frac{1}{n}\right) \times \\
& \left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{4}\right) \cdots \left(1 + \frac{1}{(n-1)}\right) \left(1 + \frac{1}{n}\right) = \\
& \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{3}{4}\right) \cdots \left(\frac{n-2}{n-1}\right) \left(\frac{n-1}{n}\right) \left(\frac{3}{2}\right) \left(\frac{4}{3}\right) \left(\frac{5}{4}\right) \cdots \left(\frac{n}{n-1}\right) \left(\frac{n+1}{n}\right) = \\
& \left(\frac{1}{n}\right) \left(\frac{n+1}{2}\right) = \frac{n+1}{2n}.
\end{aligned}$$

(b)

Proof by contradiction:

If not, there exist integers x and y such that $3x + \sqrt{2}y = 0$ and $(x \neq 0 \text{ or } y \neq 0)$.

If $x = 0$, then $\sqrt{2}y = 0$ which implies that $y = 0$;

If $y = 0$, then $3x = 0$ which implies that $x = 0$;

Hence $x \neq 0$ and $y \neq 0$. But then $\sqrt{2} = \frac{-3x}{y}$, a quotient of integers, so $\sqrt{2}$ is rational.

proof for $\sqrt{2}$ is an irrational: $\sqrt{2} = \frac{a}{b}$, where $a, b \in \mathbb{Z}$ and $\text{HCF}(a, b) = 1$ implies that $2b^2 = a^2$, hence a^2 is even, hence a is even. There exists $k \in \mathbb{Z}$, $a = 2k$.

$$2b^2 = (2k)^2 = 4k^2 \quad \text{and} \quad b^2 = 2k^2,$$

so b^2 is even and hence b is even. Contradiction.

Therefore

$$(\forall x \in \mathbb{Z})(\forall y \in \mathbb{Z})[(3x + \sqrt{2}y = 0) \implies (x = 0 \text{ and } y = 0)].$$

Question 5 solutions

Given that

$$(x + y)^5 - (x^5 + y^5) = 5xy(x + y)(x^2 + xy + y^2).$$

and

$$z^n + z^{-n} = 2 \cos n\theta.$$

$$\left(z + \frac{1}{z}\right)^5 - \left(z^5 + \frac{1}{z^5}\right) = 5z\frac{1}{z}\left(z + \frac{1}{z}\right)\left(z^2 + z\frac{1}{z} + \frac{1}{z^2}\right).$$

$$\left(z + \frac{1}{z}\right)^5 - \left(z^5 + \frac{1}{z^5}\right) = 5\left(z + \frac{1}{z}\right)\left(\left(z + \frac{1}{z}\right)^2 - 1\right)$$

$$\left(z + \frac{1}{z}\right)^5 - \left(z^5 + \frac{1}{z^5}\right) = 5\left(z + \frac{1}{z}\right)^3 - 5\left(z + \frac{1}{z}\right).$$

$$(2 \cos \theta)^5 - 2 \cos 5\theta = 5(2 \cos \theta)^3 - 5(2 \cos \theta)$$

$$32 \cos^5 \theta - 2 \cos 5\theta = 40 \cos^3 \theta - 10 \cos \theta.$$

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta.$$

$$16x^5 - 20x^3 + 5x - 1 = 0.$$

Let $x = \cos \theta$, we obtain:

$$16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta - 1 = 0.$$

which implies that

$$\cos 5\theta = 1.$$

$$\theta = \frac{2k\pi}{5}, k = 0, 1, 2, 3, 4.$$

Hence

$$1, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}.$$

are the roots of $16x^5 - 20x^3 + 5x - 1 = 0$.

$$(x - 1)(16x^4 + 16x^3 - 4x^2 - 4x + 1) =$$

$$16x^5 + 16x^4 - 4x^3 - 4x^2 + x - 16x^4 - 16x^3 + 4x^2 + 4x - 1 =$$

$$16x^5 - 20x^3 + 5x - 1 = 0.$$

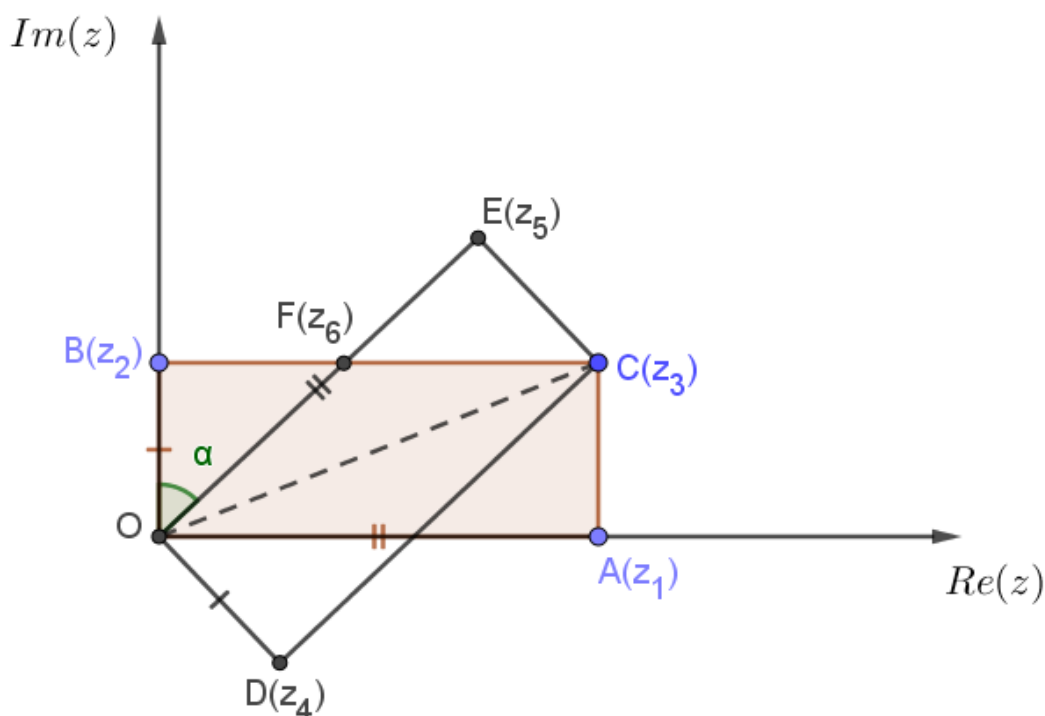
Hence

$$16x^4 + 16x^3 - 4x^2 - 4x + 1 = 0$$

has roots

$$\cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}.$$

iii. Question 6 solutions



i. $z_3 = z_1 + z_2$.

ii.

$$|z_4| = |z_2| \quad \text{and} \quad \angle BOD = \frac{\pi}{2} + \alpha.$$

$$z_2 = z_4 \operatorname{cis} \left(\frac{\pi}{2} + \alpha \right)$$

$$z_4 = \overline{z_2 \operatorname{cis} \left(\frac{\pi}{2} + \alpha \right)} = z_2 (-\sin \alpha - i \cos \alpha)$$

$$|z_5| = |z_1| \quad \text{and} \quad \angle AOE = \frac{\pi}{2} - \alpha.$$

$$z_5 = z_1 \operatorname{cis} \left(\frac{\pi}{2} - \alpha \right) = z_1 (\sin \alpha + i \cos \alpha).$$

iii.

$$z_3 = z_4 + z_5 = -z_2 (\sin \alpha + i \cos \alpha) + z_1 (\sin \alpha + i \cos \alpha) = (z_1 - z_2) (\sin \alpha + i \cos \alpha)$$

$$\overline{\operatorname{cis} \alpha} = \overline{(\cos \alpha + i \sin \alpha)} = \cos \alpha - i \sin \alpha.$$

$$i \overline{(\cos \alpha + i \sin \alpha)} = \sin \alpha + i \cos \alpha.$$

$$\text{Hence } z_3 = i(z_1 - z_2) \overline{\operatorname{cis} \alpha}$$

iv. $\triangle OBF \equiv \triangle CEF$ (AAS)

$$z_2 - z_6 = (z_5 - z_6) \operatorname{cis} \left(\frac{\pi}{2} + \alpha \right) = \left(z_1 \operatorname{cis} \left(\frac{\pi}{2} - \alpha \right) - z_6 \right) \operatorname{cis} \left(\frac{\pi}{2} + \alpha \right) = z_1 \operatorname{cis} \pi - z_6 \operatorname{cis} \left(\frac{\pi}{2} + \alpha \right)$$

$$z_2 - z_6 = -z_1 - z_6 \operatorname{cis} \left(\frac{\pi}{2} + \alpha \right)$$

$$\begin{aligned}
z_1 + z_2 &= \left(1 - \operatorname{cis}\left(\frac{\pi}{2} + \alpha\right)\right) z_6 \\
z_6 &= (z_1 + z_2) \frac{1}{1 - \cos\left(\frac{\pi}{2} + \alpha\right) - i \sin\left(\frac{\pi}{2} + \alpha\right)} \\
z_6 &= (z_1 + z_2) \frac{1}{1 + \sin \alpha - i \cos \alpha} = (z_1 + z_2) \frac{1 + \sin \alpha + i \cos \alpha}{(1 + \sin \alpha)^2 + (\cos \alpha)^2} = (z_1 + z_2) \frac{1 + \sin \alpha + i \cos \alpha}{2(1 + \sin \alpha)} \\
z_6 &= \frac{z_1 + z_2}{2} \left(1 + i \frac{\cos \alpha}{1 + \sin \alpha}\right) \\
z_6 &= \frac{z_1 + z_2}{2} \left(1 + i \frac{\cos^2 \frac{\alpha}{2} - \sin^2 \frac{\alpha}{2}}{\left(\cos^2 \frac{\alpha}{2} + \sin^2 \frac{\alpha}{2}\right)^2}\right) \\
z_6 &= \frac{z_1 + z_2}{2} \left(1 + i \frac{\cos \frac{\alpha}{2} - \sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2} + \sin \frac{\alpha}{2}}\right) \\
z_6 &= \frac{z_1 + z_2}{2} \left(1 + i \frac{\cot \frac{\alpha}{2} - 1}{\cot \frac{\alpha}{2} + 1}\right) \\
z_6 &= \frac{z_1 + z_2}{2} \left(1 + i \cot \left(\frac{\pi}{4} + \frac{\alpha}{2}\right)\right) = \frac{z_1 + z_2}{2} \left(1 + i \cot \left(\frac{\pi + 2\alpha}{4}\right)\right)
\end{aligned}$$